

PENTICTON, BC, Canada

WMO: 718890

Lat: 49.460N

Lon: 119.600W

Elev: 1130

StdP: 14.11

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.9	(c) 14.0	(d) -3.0	(e) 4.9	(f) 10.8	(g) 2.7	(h) 6.6	(i) 17.0	(j) 28.7	(k) 38.4	(l) 25.5	(m) 36.8	(n) 7.7	(o) 350	(p) 0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.7	91.5	65.4	88.2	64.5	85.0	63.2	67.3	86.7	65.7	84.3	64.3	82.3	9.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.3	81.5	73.4	58.5	76.3	72.4	56.7	71.5	72.0	32.4	86.8	31.1	84.3	30.0	82.4	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.4	20.5	18.3	4.0	97.5	5.1	2.0	0.3	98.9	-2.6	100.1	-5.5	101.2	-9.2	102.6	
			DB												
			WB	2.6	70.1	4.8	1.8	-0.9	71.4	-3.7	72.5	-6.4	73.5	-9.9	74.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	31.2	33.2	40.8	48.4	57.5	63.7	71.0	69.4	60.1	48.1	38.2	31.2		
	DBStd	15.52	7.97	7.45	6.51	5.52	6.21	5.73	5.50	5.18	5.56	6.01	7.70	7.85		
	HDD50	2496	583	471	290	94	10	0	0	2	106	358	582			
	HDD65	6095	1048	891	751	497	243	90	13	20	166	524	805	1047		
	CDD50	2312	0	0	5	47	243	412	650	601	304	47	3	0		
	CDD65	434	0	0	0	0	11	52	198	155	18	0	0	0		
	CDH74	5604	0	0	0	6	241	717	2473	1901	265	1	0	0		
	CDH80	2080	0	0	0	0	54	221	1040	709	56	0	0	0		
(14) Wind	WSAvg	7.2	9.4	7.8	7.5	6.5	6.1	5.8	6.0	5.7	5.8	7.0	9.4	9.9		
(15) Precipitation	PrecAvg	13.9	1.3	0.8	0.9	0.9	1.5	1.9	1.0	0.9	0.8	1.2	1.3	1.5		
	PrecMax	19.0	2.7	1.4	2.0	2.5	3.5	3.7	3.3	3.0	2.8	3.4	2.4	2.4		
	PrecMin	8.7	0.6	0.2	0.3	0.1	0.2	0.4	0.0	0.0	0.0	0.3	0.3	0.6		
	PrecStd	2.5	0.6	0.4	0.5	0.6	0.8	0.8	0.7	0.8	0.7	0.7	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	51.4	62.0	73.6	85.5	91.6	96.4	94.3	86.0	69.4	57.5	49.4		
		MCWB	40.6	43.2	48.4	54.9	62.1	65.2	67.3	65.8	63.6	56.3	50.7	43.8		
	2%	DB	43.9	47.8	57.9	68.2	80.4	86.0	92.8	90.1	80.7	64.8	53.8	45.5		
		MCWB	39.5	41.2	46.2	51.9	59.5	63.0	65.9	65.1	61.6	53.5	47.8	40.7		
	5%	DB	41.8	44.9	54.7	64.1	75.8	81.7	89.5	87.0	76.7	61.7	51.1	42.9		
		MCWB	38.3	39.3	44.5	49.6	57.2	61.3	65.1	64.3	59.9	51.6	46.0	38.9		
	10%	DB	39.9	42.7	51.7	60.6	71.6	77.8	86.1	84.1	73.0	59.0	48.8	40.5		
		MCWB	37.0	38.0	42.9	47.5	55.3	59.4	63.8	63.3	57.7	49.8	43.9	37.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	45.3	50.1	56.0	63.2	67.2	70.1	68.7	65.5	58.2	51.5	44.8		
		MCDB	45.4	50.0	58.9	70.9	82.1	86.3	90.6	89.2	80.8	65.7	56.5	48.2		
	2%	WB	40.1	42.1	47.5	53.1	60.6	64.4	68.0	66.8	62.9	55.4	48.7	41.5		
		MCDB	43.1	46.0	55.9	65.9	76.8	83.0	87.6	85.8	77.0	63.3	52.7	44.8		
	5%	WB	38.6	40.2	45.5	50.7	58.6	62.3	66.3	65.3	60.9	53.0	46.8	39.3		
		MCDB	41.4	44.0	53.1	61.9	73.0	78.6	85.6	83.6	74.1	59.8	50.6	42.3		
	10%	WB	37.2	38.6	43.7	48.6	56.5	60.5	64.8	63.9	59.0	50.9	44.4	37.4		
		MCDB	39.7	42.1	50.7	58.8	69.4	75.3	83.2	81.8	71.6	57.5	48.2	40.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	12.3	17.3	21.7	23.6	23.5	26.7	26.4	23.4	17.8	10.8	7.7		
		MCDBR	8.7	14.1	21.9	27.5	31.7	30.6	32.4	31.5	29.3	21.5	12.8	9.6		
		MCWBR	6.4	9.3	13.1	14.8	15.2	13.7	14.4	14.5	15.3	12.8	9.0	7.3		
	5% WB	MCDBR	8.0	12.4	19.5	25.1	28.2	27.5	29.5	28.8	26.7	19.1	11.2	8.4		
		MCWBR	6.0	8.4	12.2	14.1	13.8	12.7	13.5	13.7	14.2	11.9	8.2	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.294	0.321	0.349	0.381	0.373	0.361	0.371	0.344	0.335	0.310	0.285		
	taud		2.311	2.367	2.392	2.385	2.335	2.388	2.445	2.420	2.482	2.481	2.443	2.380		
	Ebn at Noon		238	272	283	285	279	281	283	274	272	254	233	227		
	Edh at Noon		22	26	31	35	38	36	34	33	28	24	20	18		
(44) All-Sky Solar Radiation	RadAvg		324	623	986	1454	1730	1820	2104	1790	1272	728	342	246		
	RadStd		38	64	94	110	136	161	164	131	110	90	42	29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page